

```
In [ ]: # !pip install langchain-experimental tabulate pandas openai
```

```
In [6]: import pandas as pd #파이썬 언어로 작성된 데이터를 분석 및 조작하기 위한 라이브러리

#csv 파일을 데이터프레임으로 가져오기
df = pd.read_csv('booksv_02.csv') #booksv_02.csv 파일이 위치한 경로 지정
df.head()
```

Out[6]:

	title	subtitle	authors	categories	thumbnail	description	published_year	average_rating	nun
0	Gilead	NaN	Marilynne Robinson	Fiction	http://books.google.com/books/content?id=KQZCP...	A NOVEL THAT READERS and critics have been eag...	2004	3.85	
1	Spider's Web	A Novel	Charles Osborne;Agatha Christie	Detective and mystery stories	http://books.google.com/books/content?id=gA5GP...	A new 'Christie for Christmas' - - a full-lengt...	2000	3.83	
2	The One Tree	NaN	Stephen R. Donaldson	American fiction	http://books.google.com/books/content?id=OmQaw...	Volume Two of Stephen Donaldson's acclaimed se...	1982	3.97	
3	Rage of angels	NaN	Sidney Sheldon	Fiction	http://books.google.com/books/content?id=FKo2T...	A memorable, mesmerizing heroine Jennifer -- b...	1993	3.93	
4	The Four Loves	NaN	Clive Staples Lewis	Christian life	http://books.google.com/books/content?id=XhQ5X...	Lewis' work on the nature of love divides love...	2002	4.15	

```
In [7]: import os
from dotenv import load_dotenv
load_dotenv()
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

```
In [ ]: from langchain_experimental.agents.agent_toolkits import create_pandas_dataframe_agent
from langchain_classic.chat_models import ChatOpenAI
from langchain_classic.agents.agent_types import AgentType

#에이전트 생성
agent = create_pandas_dataframe_agent(
    ChatOpenAI(temperature=0, model='gpt-4.1-mini'),
    df,          #데이터가 담긴 곳
    verbose=False, #추론 과정을 출력하지 않음
    agent_type=AgentType.OPENAI_FUNCTIONS,
    allow_dangerous_code=True,
)
```

```
In [9]: agent.run('어떤 제품의 ratings_count가 제일 높아?')
```

```
Out[9]: 'ratings_count가 가장 높은 제품은 "One Hundred Years of Solitude"이며, ratings_count는 547,207입니다.'
```

```
In [26]: agent.run('가장 최근에 출간된 책은?')
```

> Entering new AgentExecutor chain...

Invoking: `python\_repl\_ast` with `{ 'query': "df.loc[df['published\_year'].idxmax()]" }`

```
title           The Yiddish Policemen's Union
subtitle         A Novel
authors          Michael Chabon
categories       Fiction
thumbnail        http://books.google.com/books/content?id=-reD1...
description      For sixty years, Jewish refugees and their des...
published_year   2007
average_rating   3.7
num_pages        414
ratings_count    51219
Name: 69, dtype: object가장 최근에 출간된 책은 2007년에 출간된 "The Yiddish Policemen's Union"입니다. 저자는 Michael Chabon입니다.
```

> Finished chain.

Out[26]: '가장 최근에 출간된 책은 2007년에 출간된 "The Yiddish Policemen\'s Union"입니다. 저자는 Michael Chabon입니다.'

```
In [18]: df[df['ratings_count'] == df['ratings_count'].max()]
df.nlargest(1, 'ratings_count')
```

Out[18]:

	title	subtitle	authors	categories	thumbnail	description	published_year	average_rating	num_pages
168	One Hundred Years of Solitude	NaN	Gabriel Garcia Marquez	Fiction	http://books.google.com/books/content?id=pgPWO...	One of the 20th century's enduring works, One ...	2003	4.06	4



```
In [24]: df[df['published_year'] == df['published_year'].max()]
df.nlargest(1, 'published_year')
```

Out[24]:

	title	subtitle	authors	categories	thumbnail	description	published_year	average_rating	num_pages
69	The Yiddish Policemen's Union	A Novel	Michael Chabon	Fiction	http://books.google.com/books/content?id=-reD1...	For sixty years, Jewish refugees and their des...	2007	3.7	



```
In [ ]: # df[df['published_year'] == df['published_year'].max()]
# 사용했을 때 가장 최근 연도는 2007로 여러개가 출력되지만
# 추론 과정을 True로 하고 보면 모델은 df.loc[df['published_year'].idxmax()] 이러한 코드를 사용하여
# 가장 처음 등장한 인덱스 기준으로 하나만 출력한다.
```